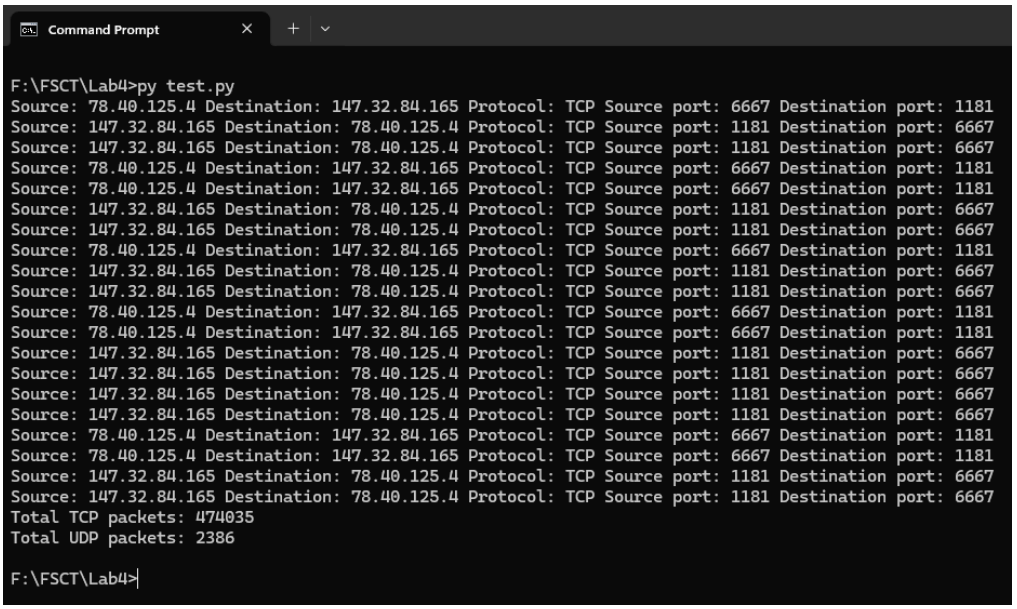
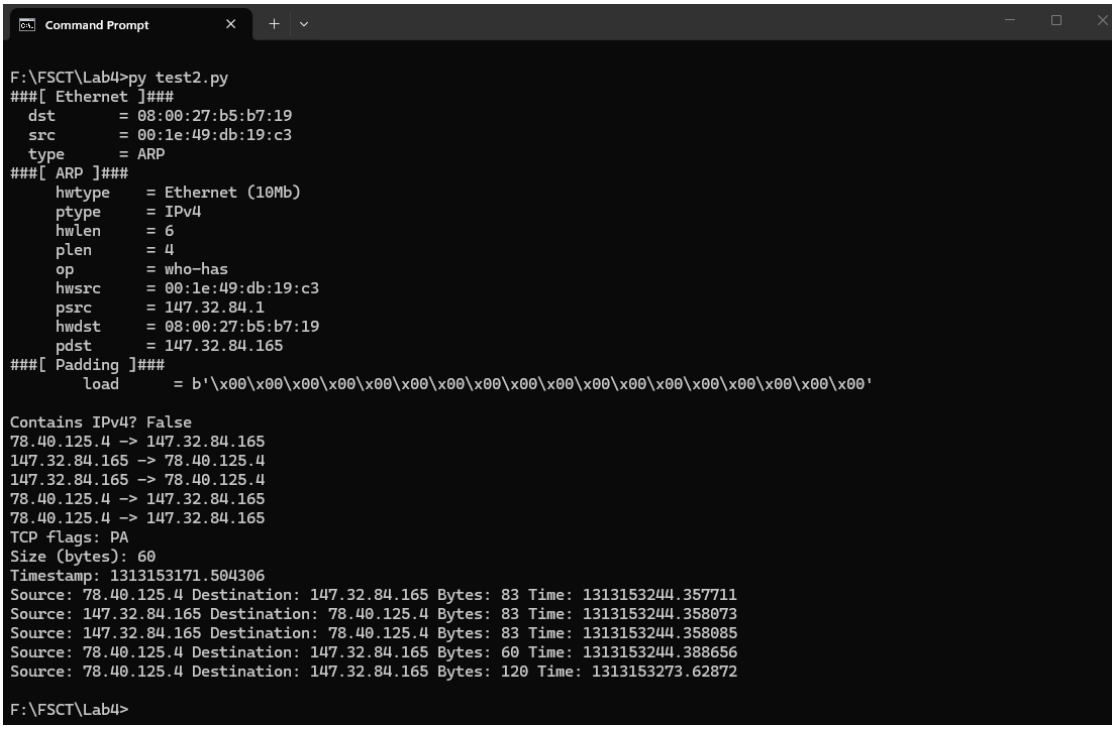
Part 1



- **How many TCP, TCP-port-80, and UDP-port-53 packets did you find?**
  TCP: 474035 and UDP: 2386
- **Why might a filter return zero packets?**
  Because there are no packets match the specified protocol or port.
- **What is the difference between filtering by protocol and filtering by port?**
  Protocol filtering selects traffic types like TCP or UDP, and port filtering selects specific services or applications.

Part 2

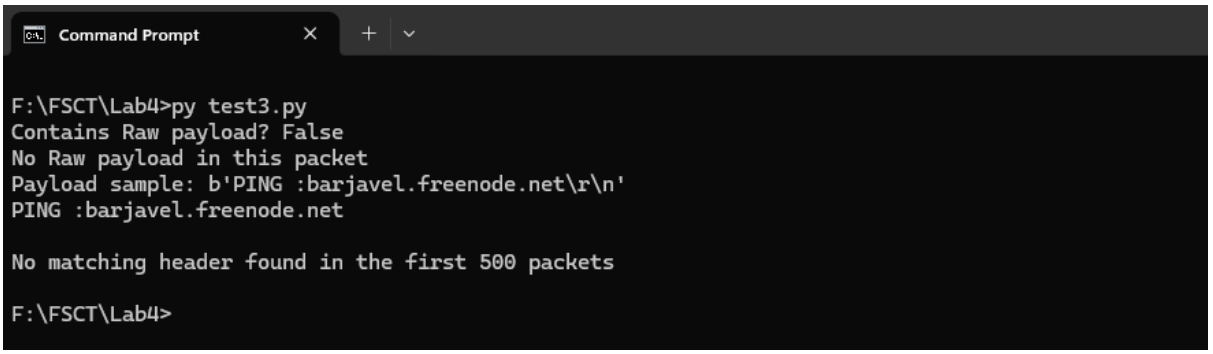


- **What do source and destination IP addresses tell you?**
  They identify where a packet originated and where it is being sent.
- **What is the purpose of TCP flags?**
  TCP flags control connections and show their status, such as SYN, ACK, and FIN.

- Why are timestamps useful when investigating possible flooding?

They help identify high packet rates within short periods

Part 3



- Did you find readable payload text?

Yes, an IRC message containing PING :barjavel.freenode.net was found.

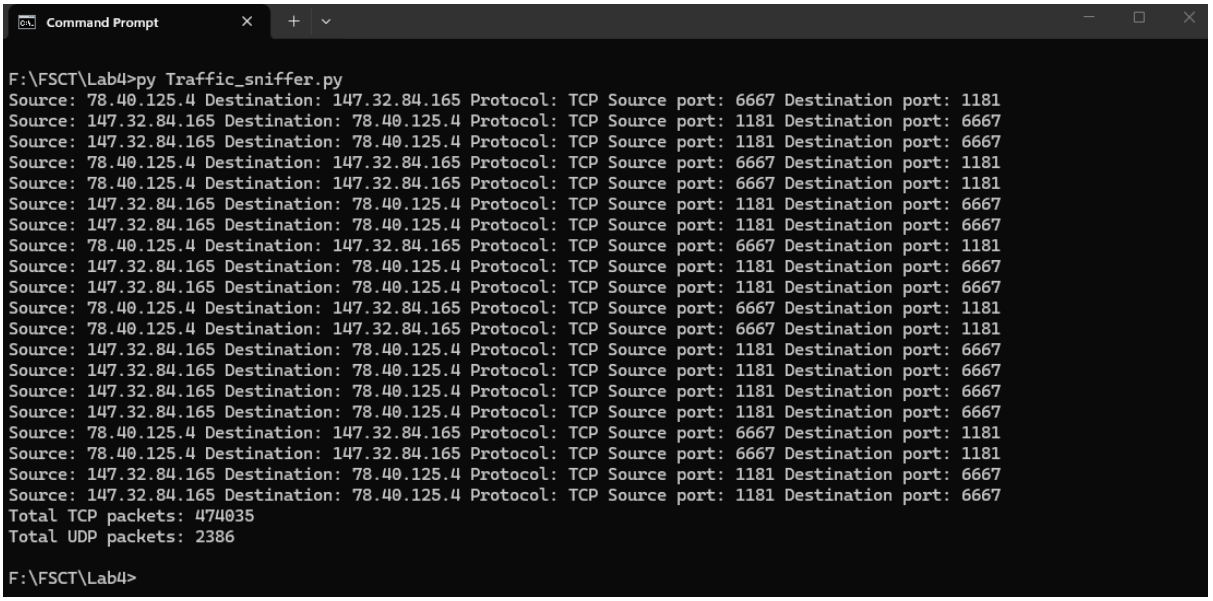
- What might plaintext HTTP expose?

Sensitive information such as usernames, passwords, cookies, and session tokens.

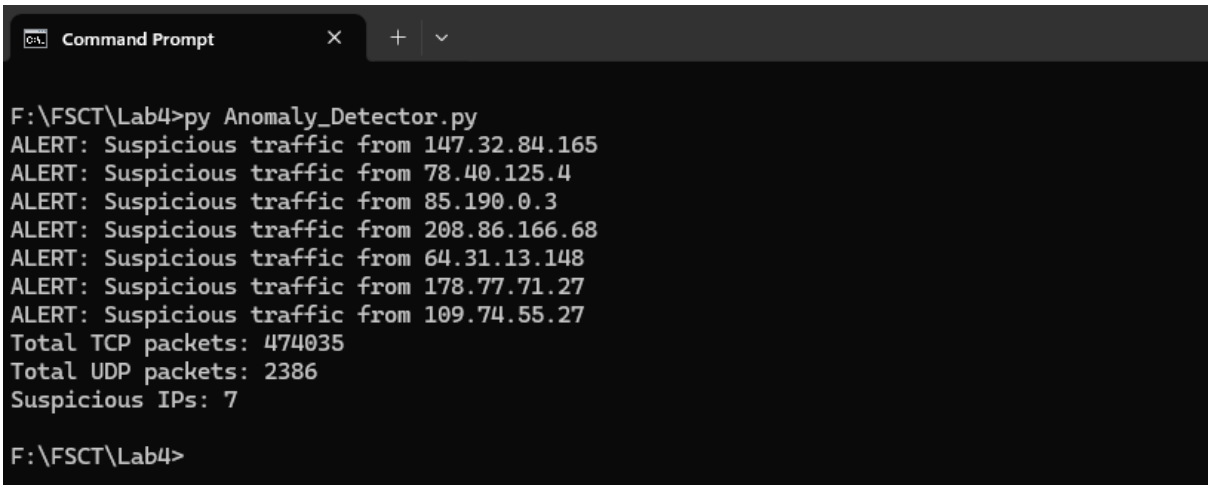
- Why can you not inspect the contents of encrypted HTTPS traffic using these simple commands?

HTTPS encrypts the data, preventing the payload from being read without decryption.

Part 4



Part 5



Part 6

1. **Why is packet inspection important for network security?**
    a. It helps identify suspicious traffic, potential attacks, and unauthorized network activity.
2. **What types of security risks can arise from unencrypted traffic?**
    a. Attackers may intercept sensitive information, steal credentials, or hijack sessions.
3. **What are the limitations of passive packet sniffing and PCAP analysis?**
    a. They only capture visible traffic and may not reveal encrypted data or activity outside the capture.
4. **How does encryption help prevent information leakage?**
    a. Encryption makes intercepted information unreadable to unauthorized users.
5. **Why might a simple packet-rate detector produce false positives?**
    a. Legitimate activities, such as downloads or large data transfers, can generate high packet rates and trigger alerts.

Part 7

In this lab, I analyzed the provided botnet PCAP file using Python and Scapy to examine network traffic and identify potentially suspicious activity. The capture included communication between IP addresses.

During payload inspection, I found a readable message containing: `**PING :barjavel.freenode.net**`. This suggests that IRC communication was present in the network capture. The message itself does not prove malicious activity, but it is worth investigating because some botnets use IRC servers to receive instructions and communicate with infected devices. The large number of TCP packets also suggests significant network activity, although packet counts alone cannot confirm an attack.

Several potential security risks were identified:

- Unencrypted network communication can expose information to unauthorized users who can capture traffic.
- Suspicious connections may also indicate malware communication or unauthorized access.

- Unusually high packet rates could indicate flooding attacks, automated scanning, or other abnormal network behaviour.

To reduce these risks, organizations should monitor network traffic regularly and investigate unusual connections. Firewalls can restrict unnecessary ports and block unauthorized communication. Intrusion detection systems can help identify suspicious traffic patterns, while encryption can protect sensitive information from interception. Keeping systems updated and using endpoint security software can also reduce the risk of malware infections.

However, packet capture analysis has limitations. Encrypted traffic prevents investigators from directly reading application data without additional information or decryption capabilities. A PCAP file also only represents traffic captured during a specific period, meaning some activity may be missing. Furthermore, the threshold-based anomaly detector used in this lab may produce false positives because legitimate applications can generate large amounts of traffic. It may also miss attacks that occur below the configured threshold.

Overall, this lab demonstrated how Scapy can be used to inspect packets, identify unusual communication patterns, and support network security investigations. Additional investigation would be required to determine whether the observed traffic was malicious.